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Student Name	Po-Han Liu
Student ID Number:	***200059267***
Supervisor's name:	***Dr.Tan Do***
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1. Introduction

Bank failure is a low-frequency but high-impact event. Even when failures are concentrated in a small number of institutions, they can reveal weaknesses in supervision, risk monitoring, and market confidence. This makes early-warning models based on bank financial ratios still relevant.

This study examines whether CAMELS-related financial ratios and Z-score measures can predict U.S. commercial bank failure, and whether the Z-score adds predictive value beyond individual CAMELS variables. CAMELS ratios capture capital adequacy, asset quality, profitability, and liquidity, while the Z-score measures distance from insolvency.

This research question is important because regulators can use early-warning models to identify vulnerable banks before failure occurs. Using FDIC Call Report data and the FDIC Failed Bank List, this study constructs a bank-year panel in which financial variables measured in year t−1 are used to predict failure in year t. A logit model is used because the dependent variable is binary.

The results show that CAMELS-related ratios are strong predictors of bank failure, broadly consistent with prior bank-failure studies. However, ln(Z-score) does not significantly improve full-sample model fit, suggesting that much of its information overlaps with its component variables. Z-score is more informative during the higher-stress 2010–2015 period.

This study contributes by testing CAMELS indicators and the Z-score in a recent U.S. commercial bank setting covering both high-failure and low-failure periods. The findings suggest that CAMELS-style ratios remain useful early-warning tools for regulators, bank managers, and market participants, but should be supplemented with liquidity, interest-rate, and duration-risk measures to capture SVB-style failures.

2. Literature Review

Three strands of literature inform this study. The first uses individual CAMELS-related ratios as direct predictors of bank failure. The second focuses on bank capital as a key driver of survival during crises. The third uses composite stability measures, such as the Z-score, to capture broader insolvency risk.

2.1 CAMELS-related ratios and U.S. bank failure prediction

Cole and White (2012) examine why U.S. commercial banks failed during the global financial crisis. Using FDIC Call Report data on approximately 7,000 to 7,600 U.S. commercial banks, they apply a logistic regression model using financial data from 2004 to 2008 as explanatory variables to predict bank failures in 2009. They find that traditional CAMELS-related indicators explain bank failure well. Banks with stronger capital, better asset quality, higher earnings, and stronger liquidity were less likely to fail, while greater exposure to commercial real estate increased failure risk. Because the model uses publicly available accounting data, it can be easily replicated and has practical value as an early-warning tool. However, their sample period ends in 2009, leaving open whether these predictors perform similarly across less stressed environments. Given the central role of capital in absorbing losses, the next section reviews studies that examine capital as a separate driver of bank survival.

2.2 Capital and bank survival during crises

Berger and Bouwman (2013) examine how bank capital affects performance during financial crises and normal times. Using Call Report data on virtually every U.S. bank from 1984Q1 to 2010Q4, they apply logit regressions to examine survival probability and OLS regressions to examine market share, separately for small, medium, and large banks. Their results show that capital improves the survival probability and market share of small banks at all times, while for medium and large banks the effect is strongest during banking crises. Their broader insight that capital matters more under stress provides the literature foundation for the time-split design tested in Section 4.2. However, their study focuses on survival and market share rather than direct failure prediction, and their sample ends in 2010. While CAMELS ratios and capital adequacy capture individual risk dimensions, composite measures provide a broader perspective on bank stability.

2.3 Z-score as a composite stability measure

Laeven and Levine (2009) investigate how ownership structure and banking regulation affect bank risk-taking. Using accounting data from 1996 to 2001 on approximately 288 large listed banks across 48 countries, they establish the Z-score, defined as (ROA+Capital-Asset Ratio (Equity/Total Assets))/σ(ROA), as a robust composite proxy for bank stability, where a higher value indicates greater distance from insolvency. Because the raw Z-score is typically right-skewed, they apply a natural logarithm transformation in their empirical analysis. A higher Z-score reflects stronger capital buffers and more stable earnings, implying a negative relationship between Z-score and bank failure probability. However, their study examines ownership and regulation rather than direct failure prediction, and their sample of large listed international banks may not generalise to FDIC-insured U.S. commercial banks. Taken together, these studies point to the research gap.

2.4 Research gap and hypothesis development

These strands all relate to bank risk. The gaps are mainly temporal, outcome-related, and scope-related. First (temporal), Cole and White (2012) and Berger and Bouwman (2013) both end in 2010, leaving the 2010–2023 period, including the 2023 SVB-era failures, largely unexamined. Second (outcome), Berger and Bouwman (2013) study survival and market share rather than next-year failure, while Laeven and Levine (2009) examine risk-taking rather than realised failure. Third (scope), Laeven and Levine (2009) rely on large listed cross-country banks, which are not representative of small and mid-sized FDIC-insured U.S. commercial banks. This study extends the literature by examining CAMELS-related ratios and the Z-score within a logit model of U.S. commercial bank failure over 2010–2023, covering both the elevated-failure 2010–2015 period and the more stable 2016–2023 period.

Combining the mechanisms outlined above, banks with stronger capital, higher profitability, and greater distance from insolvency are expected to face lower failure probability, while banks with weaker asset quality and tighter funding conditions are expected to face higher failure probability.

H1: Higher capital adequacy and profitability are negatively associated with U.S. bank failure probability, while higher non-performing loan ratios and liquidity risk are positively associated with failure probability. A higher Z-score is expected to reduce the likelihood of failure.

3. Methodology

3.1 Data and Sample

The main regression sample is based on two data sources. Financial statement data are obtained from FDIC Call Reports (Schedules RC, RC-C, RC-N, and RI) for 2009–2022 on an annual basis (FFIEC, n.d.), while failure events are identified using the FDIC Failed Bank List for 2010–2023 (FDIC, n.d.). Banks are included if they have sufficient accounting information to construct the required financial ratios and Z-score measures. This structure allows the study to estimate a one-year-ahead prediction model, in which financial variables measured in year t−1 are used to predict whether a bank fails in year t. Accordingly, the dependent variable equals one if a bank appears on the FDIC Failed Bank List in year t, and zero otherwise. Following Chiaramonte et al. (2015), the Z-score is constructed using a three-year rolling window for σ(ROA). ROA data from 2007 and 2008 are also used for the early years of the sample, although these years are not included in the main regression period.

Bank-level financial data and failure events are matched using bank identifiers. Observations with missing values for the required variables are excluded from the regression sample. In addition, the years 2018, 2021, and 2022 are excluded because no bank failures occur in those years, which would prevent the logit model from converging. The final regression sample consists of 66,733 bank-year observations, including 339 failure events. The estimation sample is also divided into two sub-periods based on bank failure years, 2010–2015 and 2016–2023. These correspond to financial variables measured in 2009–2014 and 2015–2022, respectively.

3.2 Variable Definition and Construction

The dependent variable is FAILED_NEXT_YEAR, a binary indicator equal to 1 if a bank appears on the FDIC Failed Bank List in year t, and 0 otherwise. The variable is labelled FAILED_NEXT_YEAR because it captures failure in the year following the measurement of explanatory variables.

The independent variables consist of CAMELS-related financial ratios. The CAMELS framework originally includes capital adequacy, asset quality, management, earnings, liquidity, and sensitivity to market risk. This study focuses on the CAMELS dimensions that can be directly measured using available accounting data, including capital adequacy, asset quality, profitability, and liquidity. These are measured using Capital Ratio, NPL Ratio, ROA, and the Loan-to-Deposit Ratio, respectively. In addition, Size is included as a control variable and is measured as the natural logarithm of total assets. Management and sensitivity to

market risk are excluded because these dimensions are difficult to measure consistently using publicly available accounting data.

In addition, the Z-score is included to examine whether a broader measure of bank stability improves the prediction of bank failure. It is constructed as (ROA+(Equity/Total Assets))/σ(ROA), where a higher value indicates greater distance from insolvency, and σ(ROA) is calculated using a three-year rolling window over years t−1, t−2, and t−3, which captures recent earnings volatility while preserving the sample size (Chiaramonte et al., 2015). Following Laeven and Levine (2009), the natural logarithm transformation is applied to the raw Z-score to address its right-skewed distribution.

Table 1. Variable Definitions and Expected Relationships with Bank Failure

Variable	Measurement	Expected Sign	Relevant Literature
FAILED_NEXT_YEAR	=1 if failed in t+1, 0 otherwise	Dependent Variable	Cole and White (2012)
Capital Ratio (Capital adequacy)	Equity/Total Assets	Negative (-)	Cole and White (2012); Berger and Bouwman (2013); Mayes and Stremmel (2014)
Non-performing Loans Ratio(NPL) (Asset quality)	NPL/Total Loans	Positive (+)	Cole and White (2012); Cole and Gunther (1998)
Return on Assets(ROA) (Earnings)	Net Income/Total Assets	Negative (-)	Cole and White (2012); Cole and Gunther (1998)
Loan-to-Deposit Ratio (Liquidity)	Total Loans/Total Deposits	Positive (+)	Cole and White (2012)
ln(Z-score) (Composite stability measure)	ln[(ROA + Equity/Total Assets) / σ(ROA)]	Negative (-)	Laeven and Levine (2009); Chiaramonte et al. (2015)
Size (Control)	ln(Total Assets)	Control Variable	Cole and White (2012); Berger and Bouwman (2013)

Notes: This table presents variable definitions and expected relationships with bank failure probability. Expected signs are based on theoretical predictions outlined in Section 2. NPL figures prior to 2011 are constructed using an alternative variable definition due to a change in FDIC Call Report coding (FDIC, 2011). See Appendix A for details.

3.3 Descriptive Statistics

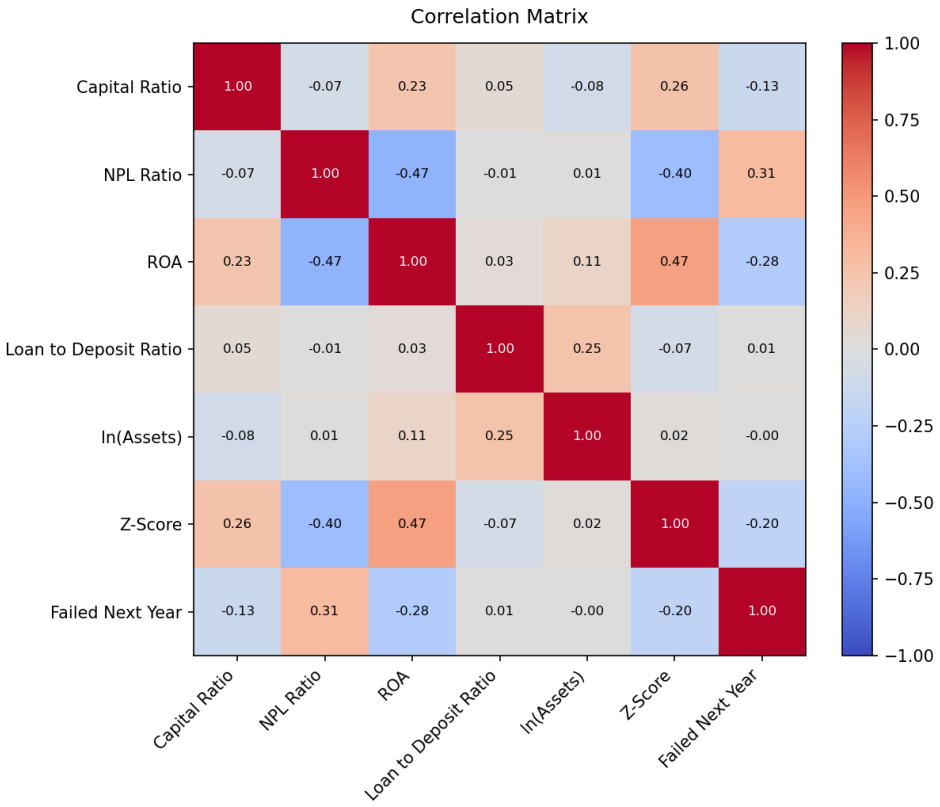
Table 2 — Descriptive Statistics								
	count	mean	std	min	25%	50%	75%	max
CAPITAL_RATIO	66733	0.111	0.045	-0.091	0.09	0.104	0.123	0.988
NPL_RATIO	66733	0.018	0.03	0	0.003	0.009	0.022	1
ROA	66733	0.007	0.011	-0.046	0.004	0.008	0.012	0.058
LOAN_DEPOSITS_RATIO	66733	0.757	0.207	0.126	0.627	0.775	0.9	1.258
SIZE	66733	12.3	1.396	7.238	11.4	12.13	12.98	21.89
ZSCORE	66733	4.03	1.249	0.222	3.358	4.133	4.825	7.58
FAILED_NEXT_YEAR	66733	0.005	0.071	0	0	0	0	1

Table 2 shows the final sample consists of 66,733 bank-year observations, including 339 failure events, indicating that bank failure is a rare outcome in the data.

The sample banks appear financially sound on average. The mean capital ratio is 11.11%, the mean NPL ratio is 1.83%, average ROA is 0.72%, and the mean loan-to-deposit ratio is 75.65%. However, the capital ratio shows substantial variation, ranging from -9.06% to 98.79%, reflecting both undercapitalised distressed banks and highly capitalised small institutions. The mean of FAILED_NEXT_YEAR is 0.0051, confirming that the sample is highly imbalanced and that bank failure is a rare event in the analysis.

Figure 1 presents the pairwise correlation matrix. The highest absolute correlation among the independent variables is 0.47, indicating that multicollinearity is not a major concern in the analysis.

Figure 1: Correlation Matrix



3.4 Model Specification

Since the dependent variable is binary, this study employs a logit model to estimate the probability of bank failure. Logit is chosen because the study focuses on one-year-ahead failure probability and coefficient interpretation, rather than time-to-failure dynamics or pure predictive performance.

$$\log\left(\frac{P(Fail_{i,t}=1)}{1-P(Fail_{i,t}=1)}\right)=\alpha+\beta_1 Capital_{i,t-1}+\beta_2 NPL_{i,t-1}+\beta_3 ROA_{i,t-1}$$

$$+\beta_4 Liquidity_{i,t-1}+\beta_5\ln(ZScore_{i,t-1})+\beta_6\ln(Assets_{i,t-1})+\delta_t$$

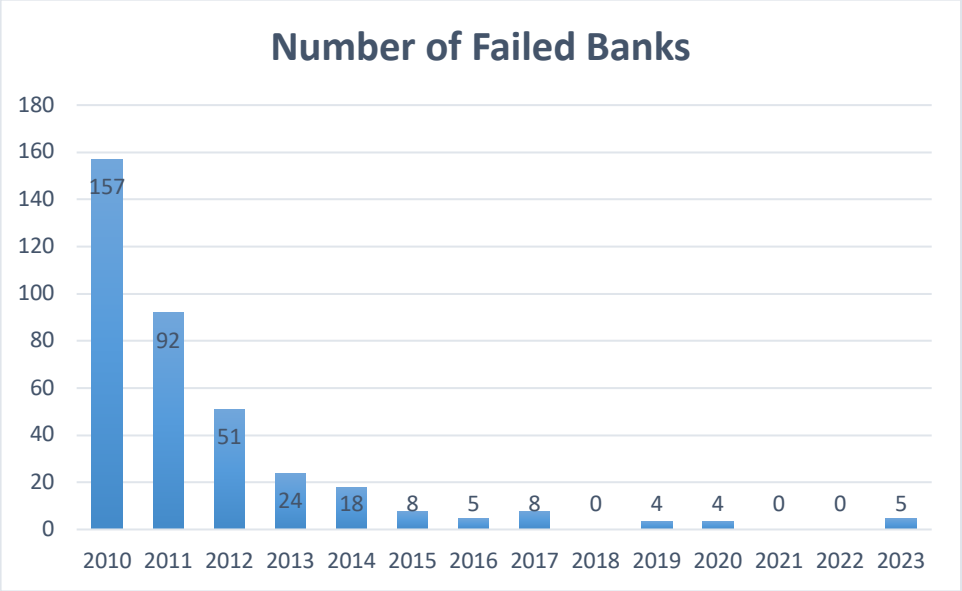
where i indexes banks, t indexes years, and δt denotes year fixed effects. All explanatory variables are measured in year $t-1$ to predict bank failure in year t. Year fixed effects are included in all specifications to control for time-specific shocks common to all banks, such as

macroeconomic conditions and regulatory changes. Standard errors are clustered at the bank level to account for within-bank correlation over time.

Models 1 and 2 follow a nested design. Model 1 includes the baseline CAMELS-related variables, while Model 2 adds ln(Z-score) to examine whether a broader measure of bank stability provides additional predictive power beyond the core CAMELS variables. The incremental contribution of ln(Z-score) is assessed using a likelihood ratio test comparing Models 1 and 2.

Model 3 uses a time-split design. The sample is divided into 2010–2015 and 2016–2023 to compare an earlier higher-failure period with a later lower-failure period.

Figure 2. Number of Failed Banks, 2010–2023



Source: FDIC Failed Bank List

This split is supported by Figure 2, which shows that bank failures are concentrated in the earlier years of the sample. It is also consistent with prior research showing that the effects of banking crises can remain visible even seven years after a crisis (Abiad et al., 2009).

Model 4 provides a robustness check. It excludes capital ratio and ROA because both are components of the Z-score. This tests whether Z-score retains predictive value when entered without its main components.

Given the highly imbalanced nature of the sample, model performance is evaluated using the Area Under the ROC Curve (AUC), which assesses the model's ability to distinguish failed from non-failed banks independently of the failure rate. To check for overfitting, five-fold stratified cross-validation is used, with stratification preserving the class balance across

folds. The optimal classification threshold is identified using Youden's J statistic (J = sensitivity + specificity − 1), which maximises the trade-off between true positive and false positive rates. Average marginal effects are also reported to make the results easier to interpret.

3.5 Method Strengths and Limitations

The logit model is well-suited for binary outcomes and produces interpretable marginal effects. Clustered standard errors at the bank level and year fixed effects improve the reliability of the estimates. Given the highly imbalanced sample, the AUC metric is used to evaluate predictive performance independently of the failure rate.

However, several limitations should be noted. Two CAMELS dimensions, management and sensitivity to market risk, are omitted due to data constraints, meaning the framework is not fully captured. This may limit the model's ability to detect failures driven primarily by interest-rate or duration-mismatch risk. Finally, as noted in Section 3.1, the exclusion of failure years with no observed failures is necessary to prevent quasi-complete separation in the logit model. This limits the generalisability of findings to periods of very low financial stress.

4. Analysis and Discussion

4.1 Regression Results

Table 3. Logit Regression Results for Bank Failure Prediction					
	M1 Baseline	**M2 +Z-Score**	**M3a 2010-15**	**M3b 2016-23**	**M4 Z-Score only**
Capital	-77.2821***	-73.4481***	-80.2427***	-46.8176***	—
	(4.6032)	(5.2097)	(6.1731)	(10.8556)	
NPL Ratio	+10.5149***	+10.3109***	+9.9700***	+14.1632***	+10.2791***
	(1.0721)	(1.0687)	(1.1605)	(2.8605)	(0.8876)
ROA	-33.7879***	-28.5718***	-15.4522*	-83.1470***	—
	(6.3643)	(7.7724)	(8.2390)	(18.3178)	
Loan/Deposit	+2.0718***	+2.0653***	+1.8478**	+1.6545	+0.5236
	(0.6373)	(0.6446)	(0.7357)	(1.1672)	(0.4587)
ln(Assets)	-0.0762	-0.0752	-0.1141	+0.2432	-0.0490
	(0.0897)	(0.0883)	(0.0797)	(0.2851)	(0.0594)
ln(Z-Score)	—	-0.1767	-0.3510*	+0.0683	-2.2707***
		(0.1401)	(0.1874)	(0.2191)	(0.1744)
N	66,733	66,733	39,603	27,130	66,733
Log-Likelihood	-665.2	-664.2	-544.9	-103.6	-864.1
McFadden R^2	0.6875	0.6880	0.7017	0.4988	0.5941
Notes: Clustered standard errors by bank in parentheses. *** p<0.01, ** p<0.05, * p<0.10. Year fixed effects included but not reported. Likelihood ratio test of Model 2 vs Model 1: χ^2 = 2.0, df = 1, p = 0.157.					

Table 4. Average Marginal Effects — Model 2			
Variable	**dy/dx**	**Std Error**	**p-value**
Capital Ratio	-0.1892***	0.0116	<0.001
NPL Ratio	+0.0266***	0.0028	<0.001
ROA	-0.0736***	0.0201	<0.001
Loan-to-Deposit Ratio	+0.0053***	0.0017	0.001
ln(Assets)	-0.0002	0.0002	0.393
ln(Z-Score)	-0.0005	0.0004	0.211
Notes: Average marginal effects (dy/dx) computed at sample means using Model 2. Clustered standard errors by bank reported in column 3. *** p<0.01, ** p<0.05, * p<0.10. Year fixed effects included but not reported.			

Table 3 reports the logit regression results for Models 1 to 4. The likelihood ratio test (χ^2 = 2.0, p = 0.157, see Table 3 notes) indicates that adding ln(Z-score) does not significantly improve model fit beyond the baseline CAMELS variables.

Overall, H1 is partially supported. CAMELS-related variables are strong and consistent predictors of bank failure, while the role of ln(Z-score) is more nuanced and is examined further in Section 4.2. Capital ratio is negative and highly significant across the models

(Model 2: -73.45***), consistent with the expected loss-absorbing role of bank capital. Table 4 shows that the average marginal effect of capital ratio is -0.1892, implying that a one-percentage-point increase in capital ratio reduces the predicted probability of failure by about 19 basis points (i.e., 0.19 percentage points). ROA is negative and significant (Model 2: -28.57***), indicating that more profitable banks are less likely to fail. The loan-to-deposit ratio is positive and significant (Model 2: +2.07***), also supporting H1. NPL ratio is positive and highly significant (Model 2: +10.31***), with an average marginal effect of +0.0266, which is consistent with H1 and with Cole and White (2012), who identify asset quality as an important predictor of bank distress. By contrast, bank size is insignificant across the models, suggesting bank size plays no independent role in predicting failure after controlling for financial condition.

However, ln(Z-score) in Model 2 is not statistically significant (p=0.207), suggesting that in the full sample it does not provide additional predictive power beyond the core CAMELS variables. One likely reason is that Z-score is mechanically constructed from profitability and capital, since it is defined as (ROA+Equity/Total Assets)/σ(ROA). However, this full-sample finding is qualified in Section 4.2, where Z-score shows differential significance across sub-periods.

4.2 Time-Split Analysis

Models 3a and 3b compare the early period (2010–2015) with the later period (2016–2023). Capital ratio remains highly significant in both periods but its coefficient is notably larger in the early period (-80.24*** vs -46.82***), suggesting that capital adequacy was a more critical predictor during the post-crisis years when many banks remained undercapitalised. ROA becomes substantially more negative in the later period (-83.15*** vs -15.45*), indicating that profitability emerged as the dominant failure predictor as the banking sector stabilised and capital levels recovered. The loan-to-deposit ratio loses statistical significance in the later period (1.65, p=0.156), suggesting that liquidity risk was less central to failure after 2016. Notably, ln(Z-score) is marginally significant in the early period (-0.35*) but insignificant in the later period (+0.07). This pattern is consistent with Berger and Bouwman (2013), suggesting that Z-score provides incremental information beyond CAMELS components only during periods of financial stress. The late period contains only 26 failures compared to 313 in the early period, so the late-period estimates should be interpreted with caution.

4.3 Robustness Check

Model 4 excludes capital ratio and ROA to test whether ln(Z-score) retains independent predictive power without its component variables. The results show that ln(Z-score) becomes highly significant (-2.27***), confirming that it does contain useful information about bank stability. Together, these results suggest that Z-score's information overlaps with its capital and profitability components, except during stress periods (Section 4.2). NPL ratio remains highly significant (+10.28***) across both Model 2 and Model 4, confirming that asset quality is the most robust predictor of bank failure regardless of model specification.

4.4 Predictive Performance

Figure 3. ROC curve

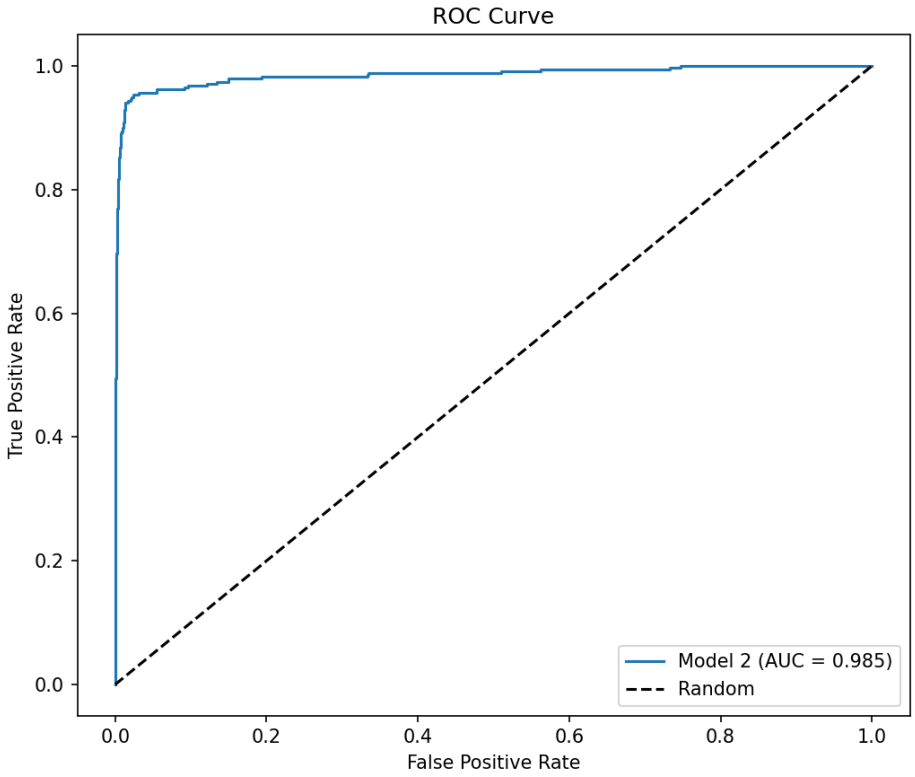


Figure 3 presents the ROC curve for Model 2. The model achieves an in-sample AUC of 0.985, which is above the random classifier baseline of 0.5. Five-fold cross-validation on the six core CAMELS and Z-score variables (without year fixed effects) yields a mean AUC of 0.984 (std = 0.006, see Appendix B), suggesting that the predictive power of the bank-specific variables is stable across folds.

At the optimal threshold identified by Youden's J statistic, the model correctly identifies 323 out of 339 failed banks (recall = 0.95), while generating 1,609 false alarms among healthy banks (precision = 0.17). This reflects a deliberate trade-off that prioritises detection over precision, as the cost of missing a genuine bank failure substantially outweighs the cost of a false alarm for regulatory purposes.

The model assigns SVB a predicted failure probability of only 0.006% based on 2022 financial data, despite its collapse in March 2023 (FDIC, 2023). SVB's CAMELS-based financial ratios appeared healthy, with a capital ratio of 7.39% and NPL ratio of only 0.21% (see Appendix C). This suggests SVB's failure was driven by interest-rate, duration, and liquidity risk factors not captured by traditional accounting ratios, highlighting the limitation of backward-looking measures in predicting sudden institutional failures (Board of Governors of the Federal Reserve System, 2023).

4.5 Implications

The findings carry three implications. For regulators, the high AUC suggests that CAMELS ratios remain useful early-warning tools, but the SVB case highlights the need to supplement them with measures of interest-rate, duration-mismatch, and liquidity risk. For bank managers, the sub-period results suggest that risk monitoring should adapt to the macroeconomic environment, with capital adequacy becoming especially important during stressed periods and profitability becoming more informative in more stable conditions. For market participants, the high-recall, low-precision trade-off means financial-ratio screens are useful as warning filters, but not as standalone failure signals.

5. Conclusion

This study examined whether CAMELS-related financial ratios and the Z-score predict U.S. commercial bank failure over 2010–2023. Using a logit model on 66,733 bank-year observations from FDIC Call Reports and the FDIC Failed Bank List, the analysis found that lower capital adequacy, weaker asset quality, lower profitability, and higher loan-to-deposit ratios significantly increase one-year-ahead failure probability. The ln(Z-score) does not improve predictive power in the full sample but becomes informative during the higher-stress 2010–2015 period.

This study contributes to the bank-failure literature by extending CAMELS-based prediction to a recent period covering both elevated-failure and stable years, and by showing that the Z-score's incremental value is conditional on the macroeconomic environment rather than uniform across periods.

The intuitive explanation is that CAMELS components and the Z-score share underlying information through capital and earnings, so the Z-score adds predictive value only when these components themselves become more volatile under stress. The SVB case further suggests that purely accounting-based measures struggle to capture interest-rate and duration-mismatch risk.

For regulators, CAMELS ratios remain valuable early-warning filters but should be supplemented with liquidity, interest-rate, and duration-risk indicators. For bank managers, capital adequacy is most critical during stress, while profitability dominates in stable periods. For market participants, the high-recall, low-precision trade-off makes ratio-based screens useful as filters rather than standalone signals.

The limitations are the omission of management and market-sensitivity dimensions, the exclusion of years with no failures, and the small number of failures after 2016. Future research could integrate market-based variables (e.g., deposit beta, unrealised securities losses) to better capture SVB-style failures.

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Tool(s) used:
ChatGPT by OpenAI and Claude by Anthropic.

Role of GenAI in this dissertation:
ChatGPT was used for grammar checking, wording refinement, and minor structure review. Claude was used for coding support, idea generation, and checking the consistency of the analysis.

Main prompts used:
The main prompts asked how the dissertation could be improved, how grammar and wording could be fixed, whether table and figure numbering was correct, and which parts of the project needed further improvement.

My contribution:
I selected the research topic, collected and checked the references, considered and evaluated AI suggestions, constructed the variables, ran the regression models, and made the final decisions on the analysis and interpretation.

Verification and critique:
I reviewed the coding process, checked the regression outputs, and read the key references in their original form. I verified the main claims, citations, and numerical results before including them in the dissertation.

Declaration:
I confirm that I am the author of this dissertation. I accept full responsibility for its accuracy, originality, and academic integrity. I verified all AI-generated content and complied with University data protection and academic integrity requirements.

Name / Student ID / Date:
Po-Han Liu / 200059267 / 1 May 2026

Appendix

Appendix A. NPL Construction and Call Report Coding

Appendix A reports the Call Report items used to construct the main variables. RCON items are used where available, while RCFD items are used as alternatives when RCON values are unavailable. Non-performing loans are defined as loans 90 days or more past due plus nonaccrual loans, based on Schedule RC-N. Pre-2011 NPL components are constructed using the closest available equivalent item codes to maintain consistency across the sample period.

Table A.1. Source Items Used for Variable Construction		
Source Items	Code	Schedule
Assets	RCON 2170 (RCFD 2170 if unavailable)	RC
Total Equity	RCON 3210 (RCFD 3210 if unavailable)	RC
Total Deposits	RCON 2200	RC
Total Loans	RCON 2122 (RCFD 2122 if unavailable)	RC-C
Net Income	RIAD 4340	RI

Table A.2. NPL Construction from Schedule RC-N			
Period	NPL Names	Status	Code
2009-2022	Commercial and Industrial Loans	90+ days past due	RCON 1607 (RCFD 1252 + RCFD1255 if unavailable)
		Nonaccrual	RCON 1608 (RCFD 1253 + RCFD1256 if unavailable)
2009-2022	All Other Loans	90+ days past due	RCON 5460 (RCFD 5460 if unavailable)
		Nonaccrual	RCON 5461 (RCFD 5461 if unavailable)
2009-2022	Credit Card Loans	90+ days past due	RCONB 576 (RCFDB 576 if unavailable)
		Nonaccrual	RCONB 577 (RCFDB 577 if unavailable)
2009-2010	Other Household/Personal Loans	90+ days past due	RCONB 579 (RCFDB 579 if unavailable)
		Nonaccrual	RCONB 580 (RCFDB 580 if unavailable)
2011-2022	Other Household/Personal Loans	90+ days past due	RCONK214+RCONK217
		Nonaccrual	RCONK215+RCONK218

Notes for Table A.2: All NPL items are sourced from Schedule RC-N. From 2011 onwards, FDIC introduced separate codes (RCONK214, RCONK215, RCONK217, RCONK218) for residential and other consumer loans, replacing the legacy aggregate codes (RCONB579, RCONB580). NPL components are constructed using the equivalent codes within each period to maintain consistency.

Appendix B. Five-Fold Cross-Validation Results

Fold 1 AUC: 0.9902
Fold 2 AUC: 0.9780
Fold 3 AUC: 0.9922
Fold 4 AUC: 0.9812
Fold 5 AUC: 0.9792
Mean 5-Fold CV AUC: 0.9841
Std: 0.0059

Notes: Cross-validation uses scikit-learn's LogisticRegression on the six core variables (Capital Ratio, NPL Ratio, ROA, Loan-to-Deposit Ratio, ln(Assets), and ln(Z-score)) with

StratifiedKFold to preserve class balance across folds. Features are standardised using StandardScaler fitted on each training fold to prevent data leakage. Year fixed effects are excluded from the cross-validation specification to focus on the predictive stability of bank-specific variables.

Appendix C. SVB Ratio Calculation

Bank	Year	Capital Ratio	NPL Ratio	Predicted Failure Probability
Silicon Valley Bank	2022	7.39%	0.21%	0.0062%

Notes: Predicted failure probability is computed using Model 2 estimates applied to SVB's 2022 financial data from the FDIC Call Report. Despite SVB's collapse in March 2023, the model assigns it a very low predicted failure probability based on its CAMELS ratios alone, illustrating the limitation of backward-looking accounting measures in capturing interest-rate and duration-mismatch risk.